CS2008: Numerical Computing
Marks -------/10 Name & R#

Quiz 3: Least Square approximations

You are given the following toy dataset:

X	Y
2	9
3	29
4	62
5	126
6	215

Steps:
1. Fit the polynomial $p(x) = a_0 + a_1x + a_2x^2$ and find $\hat{y}$.
2. Calculate the error: $\Sigma(y - \hat{y})^2$.

Solution:
As already mentioned that for m degree polynomial,

The resulting linear system is of order $m + 1$. Specifically, the coefficients a_j, $j = 0, 1, \ldots, m$ can be found by solving the linear system

$$\mathbf{Az} = \mathbf{b},$$

where $\mathbf{z} = (a_0, a_1, \ldots, a_m)^T$,

$$\mathbf{A} = \begin{pmatrix} N+1 & \sum x_i & \sum x_i^2 & \cdots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \cdots & \sum x_i^{m+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum x_i^m & \sum x_i^{m+1} & \sum x_i^{m+2} & \cdots & \sum x_i^{2m} \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \\ \vdots \\ \sum x_i^m y_i \end{pmatrix}.$$

So the Matrix Form:

$$\begin{bmatrix} 5 & 20 & 90 \\ 20 & 90 & 440 \\ 90 & 440 & 2274 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 441 \\ 2273 \\ 12179 \end{bmatrix}$$

Using a calculator, we find:

$a_0 = 12.07142857$, $a_1 = -45.67142857$, $a_2 = 53.6$

Therefore, the fitted polynomial is:

$p(x) = 12.07142857 - 45.67142857x + 53.6x^2$

We can now calculate the fitted values ($\hat{y}$):

X	Y	p(x) = ŷ
2	9	10.542857139999995
3	29	25.228571419999987
4	62	64.05714283999998
5	126	127.0285714
6	215	214.14285709999996

$\Sigma (y - \hat{y})^2 = 4.525714285714305$